



ITS
International
Symposium
Terraquorum



GEOSPHERE
2026

GUIDEBOOK

GC&C

2026



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GEOTHERMAL CASE STUDY COMPETITION GEOSPHERE 2026

A. Main Theme

The theme for the Geothermal Case Study Competition (GCSC) 2026 is **“Unlocking Subsurface Geothermal Potential to Accelerate Sustainable Energy Development”**.

B. Description

Geothermal Case Study Competition (GCSC) is a case-based subsurface energy evaluation competition under the **Geosphere Scientific Competition (GSC) 2026**, organized by the Geophysical Engineering Department, Institut Teknologi Sepuluh Nopember (ITS). The competition simulates an industry style geothermal exploration workflow focused on subsurface characterization, uncertainty reduction, and resource development decision making. The competition is conducted in two stages. In the **Preliminary Stage**, teams shall submit an abstract presenting their initial subsurface interpretation framework and analytical approach to the provided geothermal dataset. Submissions are evaluated based on conceptual clarity, technical coherence, and relevance to geothermal system assessment. Teams advancing to the **Final Stage** are required to deliver a full technical report and presentation based on integrated interpretation of geological, geophysical, geochemical, and reservoir datasets. Deliverables must include a coherent subsurface model, geothermal system characterization, resource potential assessment, uncertainty and risk evaluation, and a development strategy supported by technical justification.

Aligned with the theme **“Unlocking Subsurface Geothermal Potential to Accelerate Sustainable Energy Development,”** the competition emphasizes the challenge of converting incomplete and uncertain subsurface data into decision-ready geological models for geothermal resource evaluation. The core objective is subsurface de-risking through integrated geoscientific interpretation and systematic evaluation of geothermal prospectivity. Teams are required to integrate multidisciplinary datasets to define subsurface architecture, evaluate reservoir conditions, assess exploration risk, and rank geothermal potential. Emphasis is placed on data integration, interpretation consistency, and the ability to translate subsurface understanding into actionable development decisions.





Beyond subsurface characterization, the competition emphasizes the role of geothermal evaluation in accelerating sustainable energy development. Subsurface interpretations are expected to be directly translated into development-oriented insights that support decision-making in geothermal project planning, resource utilization, and long-term energy supply reliability. This reflects the full value chain of geothermal energy, where exploration outcomes directly influence the speed, feasibility, and scalability of sustainable energy deployment. The competition promotes a decision-oriented workflow reflecting real geothermal exploration practices, where uncertainty, trade-offs, and resource risk must be explicitly addressed within a unified subsurface framework. Participants are expected to demonstrate analytical rigor, systems thinking, and professional-level interpretation consistent with industry standards in energy resource evaluation.

C. General Rules

1. Participants must be active undergraduate students enrolled in a university or equivalent higher education institution, as proven by a valid Student Identification Card (Student ID).
2. All participants are required to complete the official registration form within the specified registration period.
3. Participants shall compete in teams consisting of a maximum of three (3) members, comprising one (1) Team Leader and up to two (2) Team Members, all from the same institution. Each participant may only be registered in one (1) team, and changes to team membership after registration closes are strictly prohibited under any circumstances.
4. The Geothermal Case Study Competition (GCSC) 2026 consists of two stages: the Preliminary Stage (Abstract Submission and Selection) and the Final Stage (Full Paper Submission and Final Presentation). The Preliminary Stage will be conducted online, while the Final Stage will be held onsite at the Department of Geophysical Engineering, Institut Teknologi Sepuluh Nopember (ITS), Surabaya, Indonesia.
5. Teams advancing to the Final Stage are required to attend the Technical Meeting. Each team must be represented by at least one (1) member. Information regarding the Technical Meeting schedule will be communicated through official channels, including email and the official competition communication group.
6. Participants shall use the software specified by the Organizing Committee for data processing and interpretation, if applicable.





7. All submissions must comply with the prescribed format, template, and submission procedures outlined in this guidebook.
8. Late submission of abstracts, full papers, or presentation materials will result in score deductions in accordance with the competition assessment policy. Modifications, revisions, or replacement of submitted files after the submission deadline are not permitted unless otherwise authorized by the Organizing Committee.
9. Any form of cheating will result in immediate disqualification.
10. All decisions made by the judging panel are final and binding.
11. Finalists must be present at the competition venue at least ten (10) minutes before the scheduled presentation session and are required to wear formal attire throughout all official competition activities.
12. All rules may be subject to change at any time based on official decisions of the organizing committee. Any updates will be communicated via email, social media, or during the Technical Meeting.

D. Case Distribution

1. Preliminary

Preliminary Case will be distributed on 01 September 2026 to the leader's email after completing registration and payment.

2. Final

The Final Case will be provided to the finalist teams and announced via the team leader's email after the completion of final registration and payment.

E. Competition Format

1. Preliminary Stage

- a. Every participant who has registered is required to carry out the preliminary stage.
- b. The Preliminary Stage will be conducted online, during which participants are required to prepare an abstract based on the provided case dataset and submit it through the designated Google Form.
- c. The abstract should align with the main theme





d. Participants must follow the specified format outlined below:

1) **Abstract layout**

- Size of page : A4
- Orientation : Portrait
- Columns : One
- Margin : 3 cm (Right, Left, Top, Bottom)
- Line spacing : 1.15
- Font type : Times New Roman

2) **Title layout**

- Font size : 16pt
- Font style : Bold
- Alignment : Centered

3) **Body text layout**

- Font size : 12pt
- Font style : Regular
- Alignment : Justified

4) **Reference** : APA style

e. Abstract template can be accessed by participants through the following link :

f. The abstract must include the following sections:

- Title
- Objectives
- Methods
- Results
- Conclusions
- Keywords (3–5 keywords in alphabetical order)

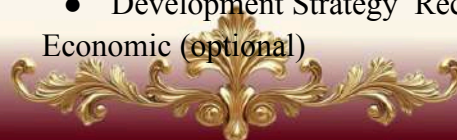
g. Participants are required to submit their abstract no later 21 September 2026, at 23:59 WIB (GMT+7). The submission must follow the file naming format : “Team Name_AbstractGCSC2026.pdf” and be uploaded via the following link:
<https://its.id/m/AbstractSubmissonGCSC2026>





2. Final Stage

- a. The top six teams with the highest scores in the Preliminary Stage will advance to the Final Stage. At this stage, all selected teams must submit a complete paper and presentation
- b. There is no restriction on the number of presentation slides. Participants are free to design and use their own presentation template.
- c. Each selected team will deliver their presentation in PPT format. The presentation session shall last for 40 minutes, comprising 5 minutes of preparation, 15 minutes of presentation, and 20 minutes of question and answer .
- d. Participants must follow the specified format outlined below:
 - 1) **Full Paper layout**
 - Size of page : A4
 - Orientation : Portrait
 - Columns : One
 - Margin : 3 cm (Right, Left, Top, Bottom)
 - Line spacing : 1.15
 - Font type : Times New Roman
 - 2) **Title layout**
 - Font size : 16pt
 - Font style : Bold
 - Alignment : Centered
 - 3) **Body text layout**
 - Font size : 12pt
 - Font style : Regular
 - Alignment : Justified
 - 4) **Reference** : APA style
- d. The full paper must contain at least the following content:
 - 1) Background
 - 2) Field Overview
 - 3) Objectives :
 - Geological Interpretation
 - Integrated Geothermal System Modeling
 - Reservoir Characterization
 - Resource Potential Assessment
 - Risk & Uncertainty Analysis
 - Development Strategy Recommendation
 - 4) Economic (optional)





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2026

5) Reference

- e. Participants are required to submit their full papers and PPT no later 15 October 2026, at 23:59 WIB (GMT+7). The submission must follow the file naming format: “Team Name_PaperGCSC2026.pdf”, “Team Name_PPTGCSC2026”, and be uploaded via the following link:
<https://its.id/m/FullPaperAndPPTSubmissionGCSC2026>

F. Assessment Criteria

1. Full Paper

- a. Originality (5%)
- b. Analysis and Interpretation (25%)
- c. Geological Aspects (15%)
- d. Technical data integration & interpretation (20%)
- e. Conceptual Geothermal System Model (20%)
- f. Resource Potential & Reservoir Evaluation (10%)
- g. Risk, Uncertainty & Development Recommendation (5%)

2. Presentation/PPT

- a. Time Management (5%)
- b. Fluency in Presentation Delivery (10%)
- c. Accuracy in Answering Questions (15%)
- d. Responsiveness in Answering (10%)

G. Timeline

1. Registration : 27 July - 31 August 2026 (3 Batch)
 - **Batch 1** : 27 July - 07 August 2026
 - **Batch 2** : 08 - 19 August 2026
 - **Batch 3** : 20 - 31 August 2026
2. Case Data Release : 01 September 2026
3. Abstract Submission : 21 September 2026
5. Finalist Announcement : 01 October 2026
6. Full Report & PPT Submission : 15 October 2026
7. Technical Meeting : 16 October 2026
8. Final Presentation : 17 October 2026





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H. The final stages of the event will be conducted offline at the following venue:

1. Final Presentation Day : Department of Geophysical Engineering ITS
2. Awarding Night : To Be Announced

I. Award

As a form of appreciation for exceptional achievements, the winners of Geothermal Case Study Competition will be granted the following awards:

- Champion : IDR 3,000,000 + Certificate
- 1st Runner Up : IDR 2,000,000 + Certificate
- 2nd Runner Up : IDR 1,500,000 + Certificate

J. Registration Guidelines

To ensure a smooth and transparent process, all participants are required to fulfill the following payment obligations based on the stage of the competition:

1. Registration Payment

Participants are required to complete the registration by paying a fee according to the registration period:

- Batch 1 : IDR 125.000 /team
- Batch 2 : IDR 150.000 /team
- Batch 3 : IDR 200.000 /team

Participants are required to submit their proof of payment for registration via the following link : <https://its.id/m/REGISTRATIONGCSC2026>

As part of the registration process, participants are required to upload the official twibbon along with the designated caption, which can be accessed via the following link: <https://its.id/m/TwibbonGCSC2026>. Participants are also required to follow GEOSPHERE's official Instagram account (@geosphereits).





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2. Bank Account Details

Payment should be made to the following bank account :

Bank JAGO 109248152122 (Account Holder by Zidan Novriza Ahmad Faraby)

K. Contact Person

If you have any questions, please contact the GEOSPHERE 2026 Geothermal Case Study Competition committee :

- Project Officer of Geothermal Case Study Competition
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